Maxwell's Equations

Electrodynamics before Maxwell

We have seen the following eqns. so far

$$\nabla . \vec{E} = \frac{1}{\varepsilon_0} \rho$$ (Gauss's law) (1)

$$\nabla . \vec{B} = 0$$ No name (2)

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$ (Faraday's law) (3)

$$\nabla \times \vec{B} = \mu_0 \vec{J}$$ (Ampère's law) (4)

$$\nabla.\vec{E} = \frac{1}{\varepsilon_0}\rho \qquad \nabla.\vec{B} = 0 \qquad \nabla\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \qquad \nabla\times\vec{B} = \mu_0\vec{J}$$

Taking divergence of (3)

$$\nabla.(\nabla\times\vec{E}) = \nabla.\left(\frac{-\partial\vec{B}}{\partial t}\right) = -\frac{\partial}{\partial t}\left(\nabla.\vec{B}\right)$$

LHS is zero because of vector property and RHS is zero because of (2)

But when we take the divergence of (4)

$$\nabla.(\nabla\times\vec{B}) = \mu_0\left(\nabla.\vec{J}\right)$$

LHS is zero because of vector property and RHS, in general, is not zero

For steady currents, **div J** is zero, but not for non-steady current and this form of Ampere's law will fail.

Yes, this is true because Ampere's law cannot hold good outside magnetostatics. Remember, It has been derived from Biot-Savarts law..

How Maxwell fixed Ampère's law?

Before going into this, we shall see the **Equation of Continuity**

Under steady state conditions, the charge density in a given region remains constant.

Let us now have a non-steady state and allow charge density to be a function of time. The net amount of charge in a closed system remains constant. This means that if the charges in a particular region decreases with time, then an equal amount of charges must appear in some other region. This transport of charges constitute a current

$$I = -\frac{dq}{dt}$$

The – sign here indicates that the charge in a specified volume decreases with time

By definition of current density $I = \oint J \bullet ds$

Therefore $$\oint J \bullet ds = -\frac{dq}{dt}$$

Ie. $$\oint J \bullet ds = -\frac{d}{dt}\int_{\tau} \rho \, d\tau$$

If we hold the surface in space, the time variation of volume integral should be solely due to time variation of ρ, hence we can write

$$\oint J \bullet ds = -\frac{\partial}{\partial t}\int_{\tau} \rho \, d\tau = -\int \frac{\partial \rho}{\partial t} d\tau$$

From Gauss theorem,

$$\oint J \bullet ds = \int \nabla \bullet J \, d\tau$$

Therefore

$$\int \nabla \bullet J \, d\tau = -\int \frac{\partial \rho}{\partial t} d\tau$$

Ie.

$$\int \left(\nabla \bullet J + \frac{\partial \rho}{\partial t} \right) d\tau = 0$$

This is true for any arbitrary finite volume, and hence

$$\nabla \bullet J + \frac{\partial \rho}{\partial t} = 0$$

This is called **Equation of Continuity** and is an expression for the fact that electric charge is conserved.

$$\nabla.\vec{E} = \frac{1}{\varepsilon_0}\rho \qquad \nabla.\vec{B} = 0 \qquad \nabla\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \qquad \nabla\times\vec{B} = \mu_0\vec{J}$$

Now let us see how Maxwell fixed Ampère's law

Continuity eqn.

From eqn.(1)

$$\nabla.\vec{J} = -\frac{\partial\rho}{\partial t} = -\frac{\partial}{\partial t}(\varepsilon_0\nabla.\vec{E}) = -\nabla.(\varepsilon_0\frac{\partial\vec{E}}{\partial t})$$

A changing electric field induces magnetic field

Maxwell added this term to the static current density and modified eqn.(4) as

$$\nabla\times\vec{B} = \mu_0\vec{J} + \mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t}$$

$$\nabla\times\vec{B} = \mu_0\vec{J} + \mu_0\vec{J}_d$$

where $\vec{J}_d \equiv \varepsilon_0\frac{\partial\vec{E}}{\partial t}$

is called the Displacement current

Thus the magnetic field is due to a conduction current J and the displacement current J_d

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

This Maxwell's eqn. is also called modified Ampere's law.
This modification does not affect anything in magnetostatics.
When E is constant, the second term is zero and this boils down to Ampere's law

In the integral form $\oint \vec{B}.dl = \mu_0 I_{enc} + \mu_0 \varepsilon_0 \int \left(\frac{\partial \vec{E}}{\partial t} \right).d\vec{a}$

Features of the displacement current $\vec{J}_d \equiv \varepsilon_0 \frac{\partial \vec{E}}{\partial t} = \frac{\partial D}{\partial t}$

1. The displacement current is a current only in a sense that it produces a magnetic field due to changing electric field. It has none other properties of current. For example, displacement current can have a finite value in vacuum where there are no charges of any type.

2. The magnitude of the displacement current is equal to the time rate of change of electric displacement vector **D**

3. The displacement current serves to make the total current continuous across discontinuities in conduction current. For example, look at the charging of a capacitor with a battery…..in the region between the plates, there is no conduction current…but there would be displacement current….and makes a closed loop (refer Griffith book for details)

4. The displacement current in a good conductor is negligible compared to conduction current except at optical frequencies (about 10^{15} Hz) - Prove…

5. The addition of displacement current to Ampere's law relates the electric field vector to the magnetic field vector. This in turn implies that in the case of time dependent fields, it is not possible to deal with electric and magnetic fields separately, but the two fields are interlinked to give rise to electromagnetic fields. Ie., the addition of Displacement current to Ampere's law results in the unification of electric and magnetic phenomena.

Maxwell's form of Ampere's law has been subjected to experimental tests and has been found valid.

Maxwell's Equations

$$\nabla . \vec{E} = \frac{1}{\varepsilon_0} \rho$$ (Gauss's law)

$$\nabla . \vec{B} = 0$$ No name

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$ (Faraday's law)

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$ (Ampere's law with Maxwell's correction)

In free space, **ρ** and **J** are zero, and hence the Maxwell eqns. become….

The four Maxwell's equations together with the force law

$$F = q\left(E + v \times B\right)$$

summarize the entire content of electrodynamics

All other equations can be derived from these equations

Example 1: derive continuity equation from Maxwell's equation

$$\nabla \bullet J + \frac{\partial \rho}{\partial t} = 0$$

Take divergence of (4) ……

Example 2: derive Coloumb's law from Maxwell's equation …….

Maxwell's equation in Matter

Inside polarized matter there will be "bound" charges and currents, over which we do not have much control.

So, it will be nice to formulate Maxwell's equations in terms of the charges we control: "free charges"

Electric polarization **P** produces the bound charge density

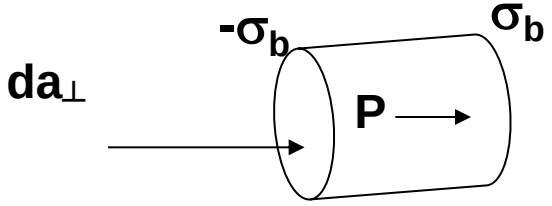
$$\rho_b = -\vec{\nabla} \cdot \vec{P}$$

Magnetic polarization **M** results in bound current

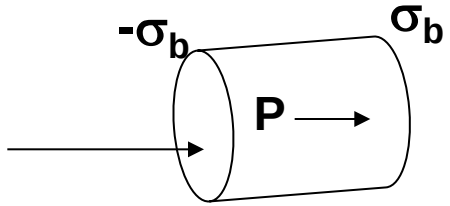
$$\vec{J}_b = \vec{\nabla} \times \vec{M}$$

For a non-static case, any change in electric polarization involves a flow of bound charge ($\mathbf{J}_p$), which must be included in the total current

Polarization introduces a charge density σ_b at one end and $-\sigma_b$ at the other end



If P is increased a bit, the charge on each end increases, giving a net current



$$dI = \frac{\partial \sigma_b}{\partial t} da_\perp = \frac{\partial P}{\partial t} da_\perp \quad \text{since} \quad \sigma_b = \vec{P} \bullet \hat{n}$$

The current density, therefore, is $\vec{J}_p = \frac{\partial \vec{P}}{\partial t}$

This is Polarization current: nothing to do with bound current

$$\vec{\nabla} \cdot \vec{J}_p = \vec{\nabla} \cdot \frac{\partial \vec{P}}{\partial t} = \frac{\partial}{\partial t}(\vec{\nabla} \cdot \vec{P}) = -\frac{\partial \rho_b}{\partial t}$$

Total charge density can be separated into two parts

$$\rho = \rho_f + \rho_b = \rho_f - \vec{\nabla} \cdot \vec{P}$$

Total current into three parts

$$\vec{J} = \vec{J}_f + \vec{J}_b + \vec{J}_p = \vec{J}_f + \hat{\nabla} \times \vec{M} + \frac{\partial \vec{P}}{\partial t}$$

Therefore Gauss law becomes $\nabla \bullet \vec{E} = \frac{1}{\varepsilon_0}\left(\rho_f - \vec{\nabla}\cdot\vec{P}\right)$ or $\vec{\nabla}\cdot\vec{D} = \rho_f$

$$\vec{D} = \varepsilon_0\vec{E} + \vec{P}$$

Ampere's law becomes

$$\vec{\nabla}\times\vec{B} = \mu_0\left(\vec{J}_f + \vec{\nabla}\times\vec{M} + \frac{\partial\vec{P}}{\partial t}\right) + \mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t}$$

or $\vec{\nabla}\times\vec{H} = \vec{J}_f + \frac{\partial\vec{D}}{\partial t}$ $\qquad \vec{H} = \frac{1}{\mu_0}\vec{B} - \vec{M}$

In terms of free charges and currents

$$\vec{\nabla}\cdot\vec{D} = \rho_f \qquad \nabla.\vec{B} = 0$$

$$\nabla\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \qquad \vec{\nabla}\times\vec{H} = \vec{J}_f + \frac{\partial\vec{D}}{\partial t}$$

Constitutive relations giving $\vec{D}$ & $\vec{H}$ in terms of $\vec{E}$ and $\vec{B}$

We saw $\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$ and $\vec{H} = \frac{1}{\mu_0}\vec{B} - \vec{M}$

We know $\vec{P} = \varepsilon_0 \chi_e \vec{E} \quad and \quad \vec{M} = \chi_m \vec{H}$

hence $\vec{D} = \varepsilon \vec{E} \quad and \quad \vec{H} = \frac{1}{\mu}\vec{B}$

where $\varepsilon \equiv \varepsilon_0 (1 + \chi_e) \quad and \quad \mu \equiv \mu_0 (1 + \chi_m)$

The Displacement current $\vec{J}_d = \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$ can be generalized as

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$$

We have seen Maxwell's equations in terms of free charges and currents

$$\vec{\nabla} \cdot \vec{D} = \rho_f \qquad \nabla . \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \qquad \vec{\nabla} \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}$$

By integrating them over an arbitrary volume or surface, we can arrive at Maxwell's equations in the integral form

Maxwell's equation in integral form

$$\oint_S \vec{D} \cdot \vec{da} = Q_{f_{enc}}$$

$$\oint_S \vec{B} \cdot \vec{da} = 0$$

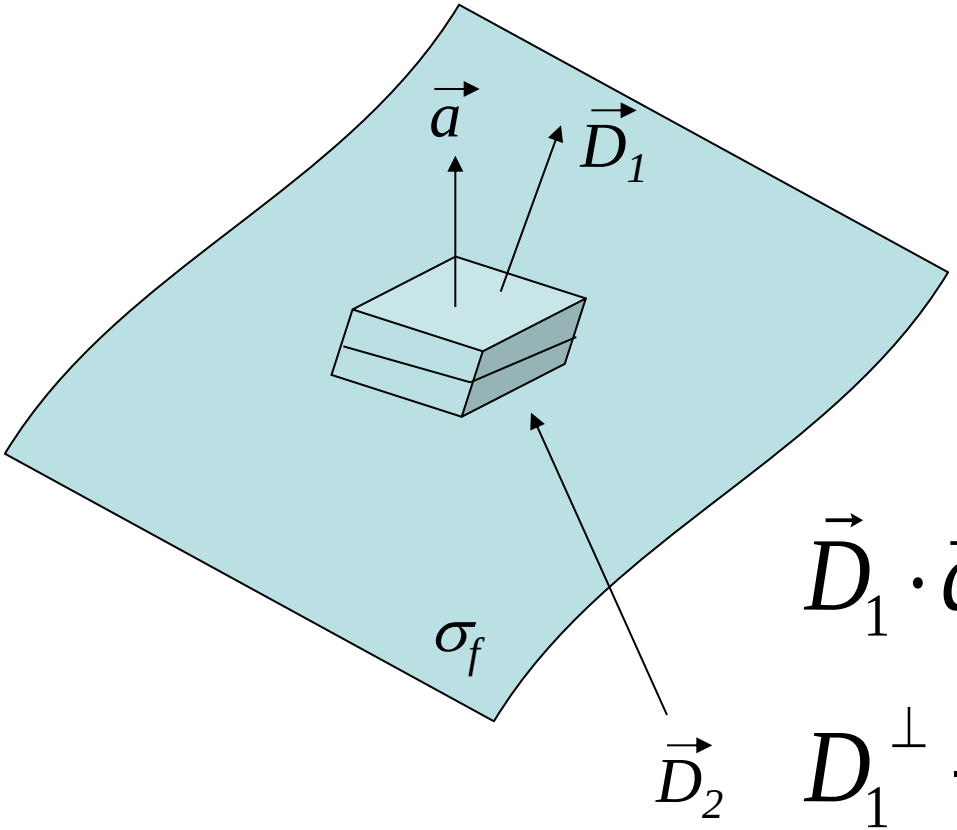
Over any closed surface S

$$\oint_p \vec{E} \cdot \vec{dl} = -\frac{d}{dt}\int_s \vec{B} \cdot \vec{da}$$

$$\oint_p \vec{H} \cdot \vec{dl} = I_{f_{enc}} + \frac{d}{dt}\int_s \vec{D} \cdot \vec{da}$$

for any surface S bounded by the closed loop p

Boundary Conditions



Applying (1) to a Wafer thin pill box

$$\vec{D}_1 \cdot \vec{a} - \vec{D}_2 \cdot \vec{a} = \sigma_f \, a$$

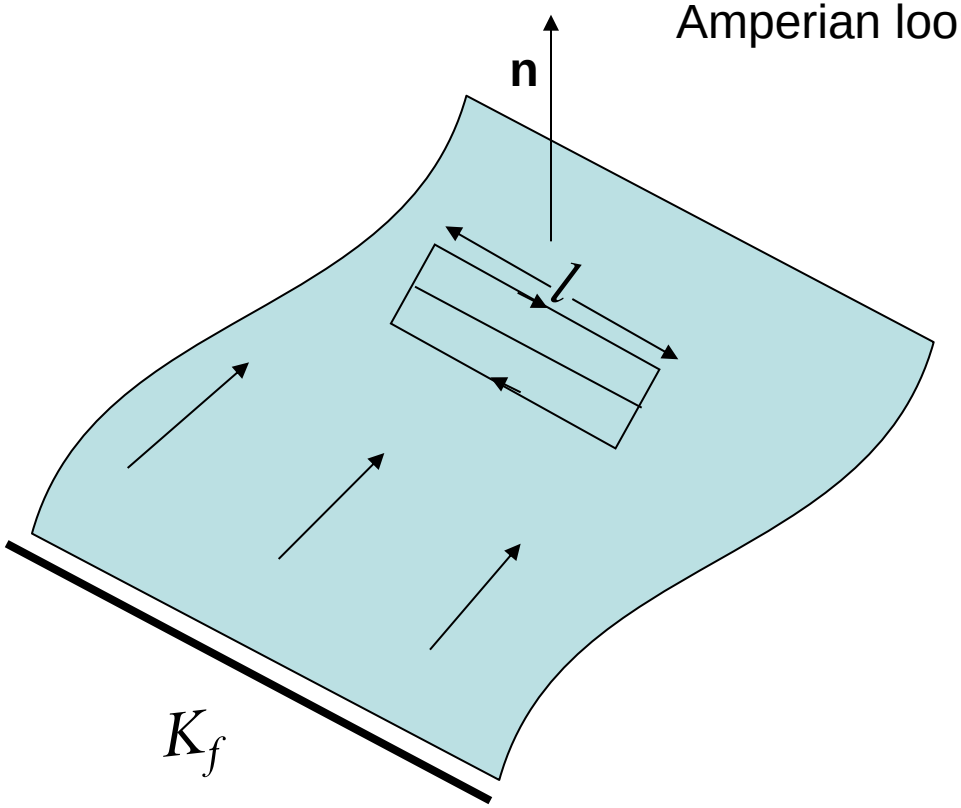
$$D_1^{\perp} - D_2^{\perp} = \sigma_f$$

Similarly

Applying (2) to the Wafer thin pill box

$$B_1^{\perp} - B_2^{\perp} = 0$$

Applying (3) and (4) to an
Amperian loop as shown if figure



$$\vec{E}_1 \cdot \vec{l} - \vec{E}_2 \cdot \vec{l} = -\frac{d}{dt}\int_s \vec{B} \cdot \overrightarrow{da}$$

$$\Rightarrow E_1'' - E_2'' = 0$$

Similarly

$$\vec{H}_1 \cdot \vec{l} - \vec{H}_2 \cdot \vec{l} = I_{f_{enc}}$$

$$I_{f_{enc}} = \vec{K}_f \cdot (\hat{n} \times \vec{l}) = (\vec{K}_f \times \hat{n}) \cdot \vec{l}$$

$$\Rightarrow \vec{H}_1'' - \vec{H}_2'' = \vec{K}_f \times \hat{n}$$

For a Linear media, they can be expressed in terms of E and B

$$\varepsilon_1 E_1^{\perp} - \varepsilon_2 E_2^{\perp} = \sigma_f \qquad \vec{E}_1'' - \vec{E}_2'' = 0$$

$$B_1^{\perp} - B_2^{\perp} = 0 \qquad \frac{1}{\mu_1}\vec{B}_1'' - \frac{1}{\mu_2}\vec{B}_2'' = \vec{K}_f \times \hat{n}$$

If there is no free charge or free current at the interface then,

$$\varepsilon_1 E_1^{\perp} - \varepsilon_2 E_2^{\perp} = 0 \qquad \vec{E}_1'' - \vec{E}_2'' = 0$$

$$B_1^{\perp} - B_2^{\perp} = 0 \qquad \frac{1}{\mu_1}\vec{B}_1'' - \frac{1}{\mu_2}\vec{B}_2'' = 0$$

Poynting's theorem

Work necessary to assemble a static charge distribution

$$W_e = \frac{\varepsilon_0}{2}\int E^2 d\tau$$

Work required to get the currents going

$$W_m = \frac{1}{2\mu_0}\int B^2 d\tau$$

Total energy stored in electromagnetic fields is

$$U_{em} = \frac{1}{2}\int\left(\varepsilon_0 E^2 + \frac{1}{\mu_0}B^2\right)d\tau$$

A given charge and current distribution gives rise to **E** and **B** (at time t).

In the next instant, dt, let us say that the charges move around a bit.

Work dW is done by the electromagnetic forces in the interval dt

According to Lenz's law, the work done on a charge q is

$$dW = \vec{F} \cdot d\vec{l} = q\left(\vec{E} + \vec{v} \times \vec{B}\right) \cdot \vec{v}dt = q\vec{E} \cdot \vec{v}dt$$

$$q = \rho d\tau \qquad\qquad \rho\vec{v} = \vec{J}$$

So the rate at which work is done on all the charges in a volume V is

$$\frac{dW}{dt} = \int_V (\vec{E} \cdot \vec{J}) d\tau$$

$\vec{E} \cdot \vec{J}$ is the work done per unit time, per unit volume or power delivered per unit volume

J can be replaced by fields using Ampere-Maxwell law

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\vec{E} \cdot \vec{J} = \frac{1}{\mu_0} \vec{E} \cdot (\vec{\nabla} \times \vec{B}) - \varepsilon_0 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

$$\vec{\nabla}\cdot(\vec{E}\times\vec{B}) = \vec{B}\cdot(\vec{\nabla}\times\vec{E}) - \vec{E}\cdot(\vec{\nabla}\times\vec{B})$$

Using Faraday's law $\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$

$$\vec{E}\cdot(\vec{\nabla}\times\vec{B}) = -\vec{B}\cdot\frac{\partial\vec{B}}{\partial t} - \vec{\nabla}\cdot(\vec{E}\times\vec{B})$$

$$\vec{B}\cdot\frac{\partial\vec{B}}{\partial t} = \frac{1}{2}\frac{\partial}{\partial t}(B^2) \qquad \vec{E}\cdot\frac{\partial\vec{E}}{\partial t} = \frac{1}{2}\frac{\partial}{\partial t}(E^2)$$

$$\vec{E}\cdot\vec{J} = \frac{1}{\mu_0}\vec{E}\cdot(\vec{\nabla}\times\vec{B}) - \varepsilon_0\vec{E}\cdot\frac{\partial\vec{E}}{\partial t}$$

$$= -\frac{1}{2}\frac{\partial}{\partial t}(\varepsilon_0 E^2 + \frac{1}{\mu_0}B^2) - \frac{1}{\mu_0}\vec{\nabla}\cdot(\vec{E}\times\vec{B})$$

$$\frac{dW}{dt} = \int_V (\vec{E} \cdot \vec{J}) d\tau$$

$$\frac{dW}{dt} = -\frac{d}{dt}\int_V \frac{1}{2}(\varepsilon_0 E^2 + \frac{1}{\mu_0} B^2) d\tau - \frac{1}{\mu_0}\oint_S (\vec{E} \times \vec{B}) \cdot d\vec{a}$$

Poynting's theorem- Work energy theorem in electrodynamics

The work done on the charges by the electromagnetic fields is equal to the decrease in energy stored in the field, less the energy that flowed out through the surface

The energy per unit time per unit area transported by the fields is called the **Poynting vector**

$$\vec{S} = \frac{1}{\mu_0}(\vec{E} \times \vec{B})$$

$\vec{S} \cdot d\vec{a}$ is the energy flux or energy flux density

$$\frac{dW}{dt} = -\frac{dU_{em}}{dt} - \oint_S \vec{S} \cdot d\vec{a}$$

The work W done on charges will increase their mechanical energy (kinetic, potential.). If we let u_{mech} denote the mechanical energy density

$$\frac{dW}{dt} = \frac{d}{dt}\int_V u_{mech}\, d\tau$$

U_{em} is the energy density of the fields

$$u_{em} = \frac{1}{2}(\varepsilon_0 E^2 + \frac{1}{\mu_0} B^2)$$

$$\frac{d}{dt}\int_V (u_{mech} + u_{em})d\tau = -\oint_S \vec{S} \cdot d\vec{a} = -\int_V (\vec{\nabla} \cdot \vec{S})d\tau$$

hence

$$\frac{\partial}{\partial t}(u_{mech} + u_{em}) = -\vec{\nabla} \cdot \vec{S}$$

Differential version of Poynting's theorem

This is similar to the continuity equation

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \vec{J}$$

Poynting's theorem describes the energy flow in the same way as continuity equation describes the flow of charge.

EM Waves in Vacuum

In terms of free charges and currents, Maxwell's equations are

$$\vec{\nabla} \cdot \vec{D} = \rho_f \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla . \vec{B} = 0 \qquad \vec{\nabla} \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}$$

When there are no free charges or free currents, Maxwell's eqns become

$$\vec{\nabla} \cdot \vec{E} = 0 \quad (1) \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (3)$$

$$\nabla . \vec{B} = 0 \quad (2) \qquad \vec{\nabla} \times \vec{B} = \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \quad (4)$$

By taking curl of equation (3)

$$\nabla \times \nabla \times \vec{E} = \nabla \times (-\tfrac{\partial \vec{B}}{\partial t})$$

$$\nabla\left(\nabla \bullet \vec{E}\right) - \nabla^2 \vec{E} = -\frac{\partial}{\partial t}\left(\nabla \times \vec{B}\right)$$

$$\nabla^2 \vec{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

since $\vec{\nabla} \cdot \vec{E} = 0$

$$\vec{\nabla} \times \vec{B} = \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

SImilarly taking curl of eqn (4), we can get

$$\nabla^2 \vec{B} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

Thus we have decoupled the E and B equations as

$$\nabla^2 \vec{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \qquad \nabla^2 \vec{B} = \mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

These equations, known as wave equations, can be written in a more compact form using wave operator $\Box$

$$\Box^2 = \nabla^2 - \mu_0 \varepsilon_0 \frac{\partial^2}{\partial t^2}$$

Then

$$\Box^2 \vec{E} = 0$$

$$\Box^2 \vec{B} = 0$$

The cost we have paid in decoupling is that we get into 2nd order eqns.

Now we will try the simplest solution to this wave equation namely The Plane Monochromatic electromagnetic waves

$$\vec{E}(\vec{r},t) = \left|E_0\right| \hat{n} \cos(\vec{k}.\vec{r} - \omega t + \delta)$$

This corresponds to sinusoidal waves propagating in a direction specified by wave vector **K**. The wave number k, which is the magnitude of wave vector **K,** is related with wavelength $\lambda = 2\pi/k$.

The unit vector $\hat{n}$ specifies the direction of electric field and also called the polarization of the plane waves, E_0 is the amplitude of the wave and real constant δ is arbitrary phase angle and frequency $\nu = \omega/2\pi$. The magnetic field corresponding to EM waves has the same frequency and wave vector

$$\vec{B}(\vec{r},t) = \left|B_0\right| \hat{n}' \cos(\vec{k}.\vec{r} - \omega t + \delta')$$

Plane EM waves

Wavefronts which is nothing but surface of equal phases is in this case is given by

$$\vec{k}.\vec{r} - \omega t + \delta = \text{Constant} \qquad \Rightarrow \vec{k}.\vec{r} - \omega t = \text{Constant}$$

At any given instant of time the shape of the wavefronts is given by

$$\vec{k}.\vec{r} = \text{Constant}$$

which is nothing but a plane in space with has normal in the direction of wave vector **K.** The fact that the wave forms above has unique wavelength is the reason why it is called "Monochromatic"

Wave packets

What if the EM waves are wave packets comprising many wavelengths (and hence many frequencies) as well as wave fronts that are not planes but more complex object like sphere, elliptic or cylinders?

Answer lies in superposition principle. The Maxwell's eqns. are linear equations and solutions corresponding to different wavelength, direction, and amplitude can be superimposed to form general wave packets of requisite properties. Conversely a given wave packet is decomposed in terms of plane monochromatic waves.

$$\vec{E}(\vec{r},t) = \left|E_0\right| \hat{n} \cos(\vec{k}.\vec{r} - \omega t + \delta) \Rightarrow \vec{E}(\vec{r},t) = \mathrm{Re}(E_0 \hat{n} e^{i(\vec{k}.\vec{r} - \omega t)})$$

$$\vec{B}(\vec{r},t) = \left|B_0\right| \hat{n}' \cos(\vec{k}.\vec{r} - \omega t + \delta') \Rightarrow \vec{B}(z,t) = \mathrm{Re}(B_0 \hat{n}' e^{i(\vec{k}.\vec{r} - \omega t)})$$

Where E_0 and B_0 are complex numbers given as

$$E_0 = |E_0| e^{i\delta} \qquad B_0 = |B_0| e^{i\delta'}$$

It is clear that we could just have taken imaginary part instead of real part in the above and that as well would be a solutions to the linear ME or a linear combinations of the two would also be solutions. Therefore the general solutions can be in the form

$$\vec{E}(\vec{r},t) = E_0 \hat{n} e^{i(\vec{k}.\vec{r}-\omega t)} \qquad \vec{B}(\vec{r},t) = B_0 \hat{n}' e^{i(\vec{k}.\vec{r}-\omega t)}$$

Here we take real parts of the solutions to obtain the physical field. The advantage in writing in exponential form that the ME which is partial differential equations for the fields becomes ordinary algebraic equations for the amplitudes. This happens essentially because derivative of exponential is again a exponential.

$$\frac{\partial}{\partial t} e^{-i\omega t} = -i\omega e^{-i\omega t}$$

$$\nabla e^{i\vec{k}.\vec{r}} = i\vec{k} e^{i\vec{k}.\vec{r}} \qquad \nabla.(\hat{n} e^{i\vec{k}.\vec{r}}) = (i\vec{k}.\hat{n}) e^{i\vec{k}.\vec{r}}$$

$$\nabla \times (\hat{n} e^{i\vec{k}.\vec{r}}) = (i\vec{k} \times \hat{n}) e^{i\vec{k}.\vec{r}}$$

Therefore we can effectively use the following replacements in Maxwell's equations when dealing plane wave solutions.

$$\nabla \rightarrow i\vec{k} \qquad \frac{\partial}{\partial t} \rightarrow -i\omega$$

Now use these replacements in the wave equations

$$\nabla^2\vec{E} = \mu_0\varepsilon_0 \frac{\partial^2\vec{E}}{\partial t^2} \qquad \nabla^2\vec{B} = \mu_0\varepsilon_0 \frac{\partial^2\vec{B}}{\partial t^2}$$

$$\mu_0\varepsilon_0\omega^2 - k^2 = 0 \qquad \omega = ck \text{ where } c \equiv \frac{1}{\sqrt{\mu_0\varepsilon_0}}$$

$$\text{or } \omega/k \equiv \nu\lambda \equiv c$$

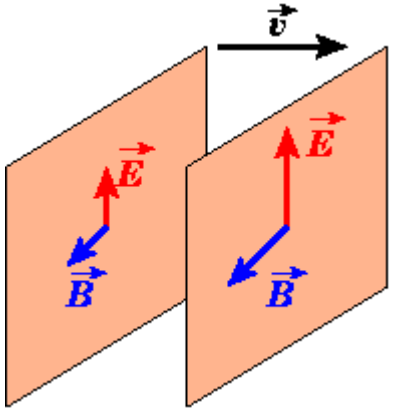
Points relating to EM waves

1. EM waves propagates in free space with velocity c.

Now let us substitute the plane wave forms in divergence equations, ie, $\nabla \cdot \vec{E}$ *and* $\nabla \cdot \vec{B}$ we get

$$\vec{k}.\hat{n} = 0 \qquad \vec{k}.\hat{n}' = 0$$

2. EM waves are transverse waves: both the electric field and the magnetic field of such wave are normal to the direction of propagation of the waves.



Now let us substitute the plane wave forms in curl equations

$$E_0(\vec{k}\times\hat{n}) - \omega B_0 \hat{n}' = 0 \quad \Rightarrow \frac{E_0(\hat{k}\times\hat{n})}{c} = B_0\hat{n}'$$

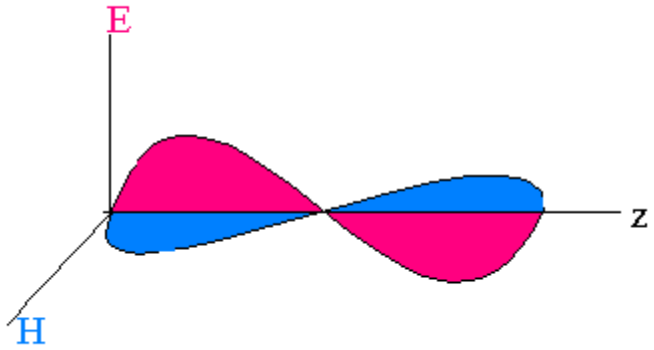
$$B_0(\vec{k}\times\hat{n}') + \omega\mu_0\varepsilon_0 E_0\hat{n} = 0 \quad \Rightarrow B_0 c(\hat{n}'\times\hat{k}) = E_0\hat{n}$$

3. Electric and magnetic field are mutually orthogonal

4. **E**, **B** and **K** form a right handed triad of vectors in such a wave, at all times.

5. The phases of the electric and magnetic in a plane EM waves are the same.

6. The amplitude of electric and magnetic field in plane EM waves are not independent, but are related to each other



Summarizing we can write

$$\vec{E}(\vec{r},t) = |E_0|\,\hat{n}\cos(\vec{k}.\vec{r} - \omega t + \delta)$$

$$\vec{B}(\vec{r},t) = |E_0|\,\frac{(\hat{k}\times\hat{n})}{c}\cos(\vec{k}.\vec{r} - \omega t + \delta)$$

Represents plane Monochromatic wave solutions to the source free ME in vacuum with

$$\hat{k}.\hat{n} = 0 \qquad \omega = ck$$ And

- Wave vector **K** gives direction of propagation and also wavelength.
- The polarization $\hat{n}$ which gives direction of electric field
- The complex number E_0, which specifies the amplitude and the phase of the electric field.

All other quantities can be deduced from the above

Energy and momentum density in a plane wave

$$U_{em} = \frac{1}{2}\int\left(\varepsilon_0 E^2 + \frac{1}{\mu_0}B^2\right)d\tau$$

For the plane electromagnetic waves

$$B^2 = \frac{1}{c^2}E^2 \quad \Rightarrow B^2 = \mu_0\varepsilon_0 E^2 \qquad \Rightarrow U_{em} = \varepsilon_0 E^2 \quad = \varepsilon_0 E^2{}_0 \cos^2(\vec{k}.\vec{r} - \omega t + \delta)$$

Contributions from electric and magnetic fields are equal

Energy flux density in a plane wave

$$\vec{S} = \frac{1}{\mu_0}(\vec{E}\times\vec{B}) = c\varepsilon_0 E^2{}_0 \cos^2(\vec{k}.\vec{r} - \omega t + \delta)\hat{e}_z = cU_{em}\hat{e}_z$$

For, in a time δt, a length $c\delta t$ passes through area A, carrying with it a energy $Ac\delta t\, U_{em}$. The energy per unit time per unit area is therefore cU_{em}

EM field not only carry energy but also momentum, the momentum density stored in the field is given by

$$\vec{\wp} = \frac{1}{c^2}\vec{S} \quad = \frac{1}{c}U_{em}\hat{e}_z$$

In the case of light, the wavelength is so short and period is so brief that a macroscopic measurement encompass many cycles, so all we want is average value

$$U_{em} = \varepsilon_0 E^2{}_0 \cos^2(\vec{k}.\vec{r} - \omega t + \delta) \qquad \Rightarrow \langle U_{em} \rangle = \frac{1}{2}\varepsilon_0 E_0{}^2$$

$$\Rightarrow \langle \vec{S} \rangle = \frac{1}{2} c\varepsilon_0 E_0{}^2 \hat{e}_z = I \text{ (Intensity)} \qquad \langle \vec{\wp} \rangle = \frac{1}{2c}\varepsilon_0 E_0{}^2 \hat{e}_z$$

Plane EM waves in Matter

In a source free region, i.e. where no free charges and currents are there, the ME can become

$$\vec{\nabla}\cdot\vec{D}=0 \qquad \nabla.\vec{B}=0$$

$$\nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t} \qquad \vec{\nabla}\times\vec{H}=\frac{\partial\vec{D}}{\partial t}$$

If the medium is linear then

$$\vec{D}=\varepsilon\vec{E} \qquad and \qquad \vec{H}=\frac{1}{\mu}B$$

And if the material is homogeneous then the permittivity and permeability does not vary from point to point

$$\vec{\nabla}\cdot\vec{E}=0 \qquad \nabla.\vec{B}=0$$

$$\nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t} \qquad \vec{\nabla}\times\vec{B}=\mu\varepsilon\frac{\partial\vec{E}}{\partial t}$$

This differs from vacuum analogs only in fourth equation

Therefore EM waves travels in matter with velocity

$$v \equiv \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{n} \qquad n = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}}$$

Here n is known as index of refraction, for most materials permeability is very close to vacuum value, therefore

$$n = \sqrt{\frac{\varepsilon}{\varepsilon_0}} = \varepsilon_r$$

Since relative permittivity is always greater than one, therefore, the EM waves travels in slower in matter than in Vacuum

Reflection, and refraction of Plane waves at Normal incidence

Incident light

$$\vec{E}_I(z,t) = \tilde{E}_{0I}\hat{x}e^{i(k_1 z-\omega t)}$$

$$\vec{B}_I(z,t) = \hat{y}\frac{\tilde{E}_{0I}}{v_1}e^{i(k_1 z-\omega t)}$$

Reflected light

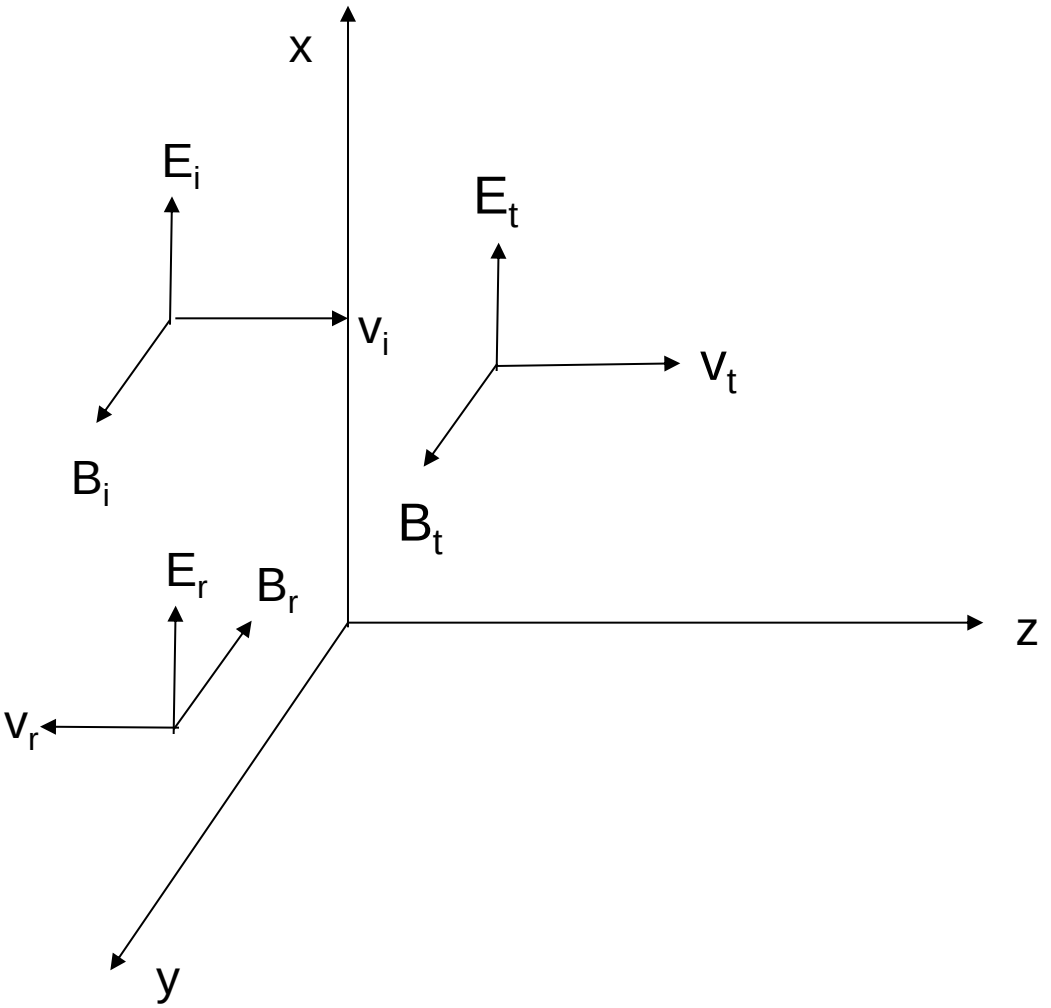
$$\vec{E}_R(z,t) = \tilde{E}_{0R}\hat{x}e^{i(-k_1 z-\omega t)}$$

$$\vec{B}_R(z,t) = -\hat{y}\frac{\tilde{E}_{0R}}{v_1}e^{i(-k_1 z-\omega t)}$$

Transmitted light

$$\vec{E}_T(z,t) = \tilde{E}_{0R}\hat{x}e^{i(k_2 z-\omega t)}$$

$$\vec{B}_T(z,t) = \hat{y}\frac{\tilde{E}_{0T}}{v_2}e^{i(k_2 z-\omega t)}$$



At the interface they should follow the boundary condition

$$\varepsilon_1 E_1^{\perp} - \varepsilon_2 E_2^{\perp} = 0 \qquad \vec{E}_1^{\parallel} - \vec{E}_2^{\parallel} = 0$$

$$B_1^{\perp} - B_2^{\perp} = 0 \qquad \frac{1}{\mu_1}\vec{B}_1^{\parallel} - \frac{1}{\mu_2}\vec{B}_2^{\parallel} = 0$$

There is no component perpendicular to the plane so equation 1 and 2 are obvious. From 3 and 4 we get

$$\tilde{E}_{0I} + \tilde{E}_{0R} = \tilde{E}_{0T}$$

$$\frac{1}{\mu_1}\left(\frac{1}{v_1}\tilde{E}_{0I} - \frac{1}{v_1}\tilde{E}_{0R}\right) = \frac{1}{\mu_2}\left(\frac{1}{v_2}\tilde{E}_{0T}\right) = 0$$

$$\tilde{E}_{0I} - \tilde{E}_{0R} = \beta\tilde{E}_{0T}$$

Where $\beta = \dfrac{\mu_1 v_1}{\mu_2 v_2} = \dfrac{\mu_1 n_2}{\mu_2 n_1}$

These equation can be solved for getting amplitude of reflected and transmitted light in terms of incident light

$$\tilde{E}_{0R} = \frac{1-\beta}{1+\beta}\tilde{E}_{0I} \qquad \tilde{E}_{0T} = \frac{2}{1+\beta}\tilde{E}_{0I}$$

If the permittivity is taken closer to the vacuum value then

$$\beta = \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{v_1}{v_2}$$

Therefore

$$\tilde{E}_{0R} = \frac{v_2 - v_1}{v_2 + v_1}\tilde{E}_{0I} \qquad \tilde{E}_{0T} = \frac{2v_2}{v_2 + v_1}\tilde{E}_{0I}$$

$$E_{0R} = \left| \frac{n_1 - n_2}{n_2 + n_1} \right| E_{0I} \qquad E_{0T} = \frac{2n_1}{n_2 + n_1} E_{0I}$$

Now intensity of the light is given as $\frac{1}{2} v \varepsilon E_0^{\,2} = I$

Therefore ratio of the reflected intensity to incident intensity is

$$R = \frac{I_R}{I_I} = \left(\frac{E_{0R}}{E_{0I}} \right)^2 = \left(\frac{n_1 - n_2}{n_2 + n_1} \right)^2$$

Similarly ratio of the transmitted intensity to incident intensity is

$$T = \frac{I_T}{I_I} = \frac{\varepsilon_2 v_2}{\varepsilon_1 v_1} \left(\frac{E_{0T}}{E_{0I}} \right)^2 = \frac{4 n_2 n_1}{\left(n_2 + n_1 \right)^2}$$

And we can very well see that $R + T = 1$

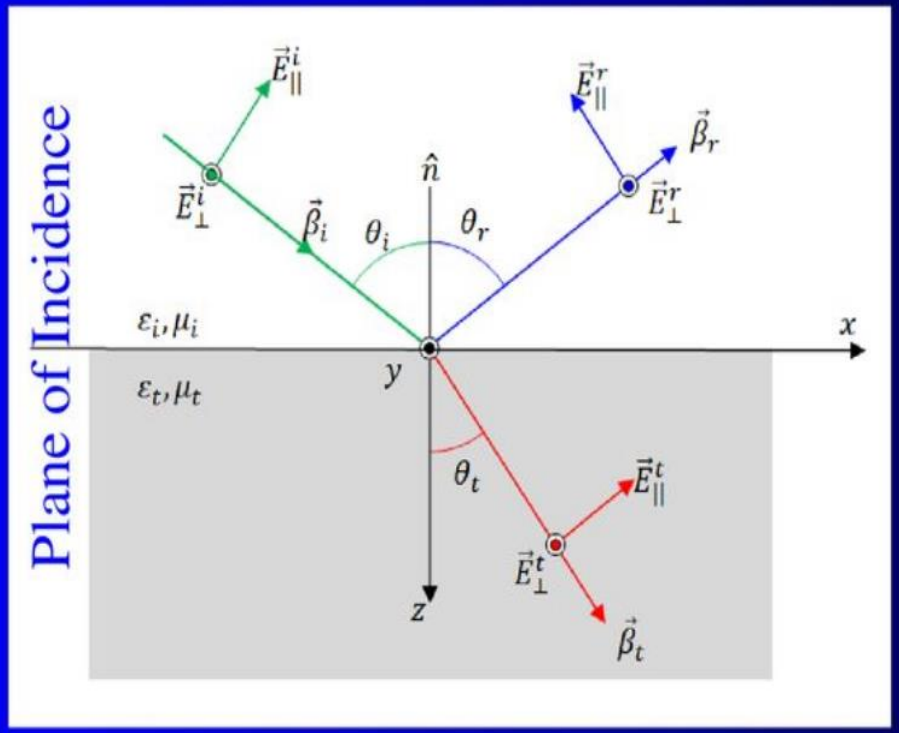
Plane EM Wave: Oblique Incidence

$$\vec{E}_{\perp}^{i} = \vec{E}_{0\perp}^{i} e^{-j\vec{\beta}_i \cdot \vec{r}}$$

$$\vec{E}_{||}^{i} = \vec{E}_{0||}^{i} e^{-j\vec{\beta}_i \cdot \vec{r}}$$

$$\vec{E}_{\perp}^{r} = \vec{E}_{0\perp}^{r} e^{-j\vec{\beta}_r \cdot \vec{r}}$$

$$\vec{E}_{||}^{r} = \vec{E}_{0||}^{r} e^{-j\vec{\beta}_r \cdot \vec{r}}$$



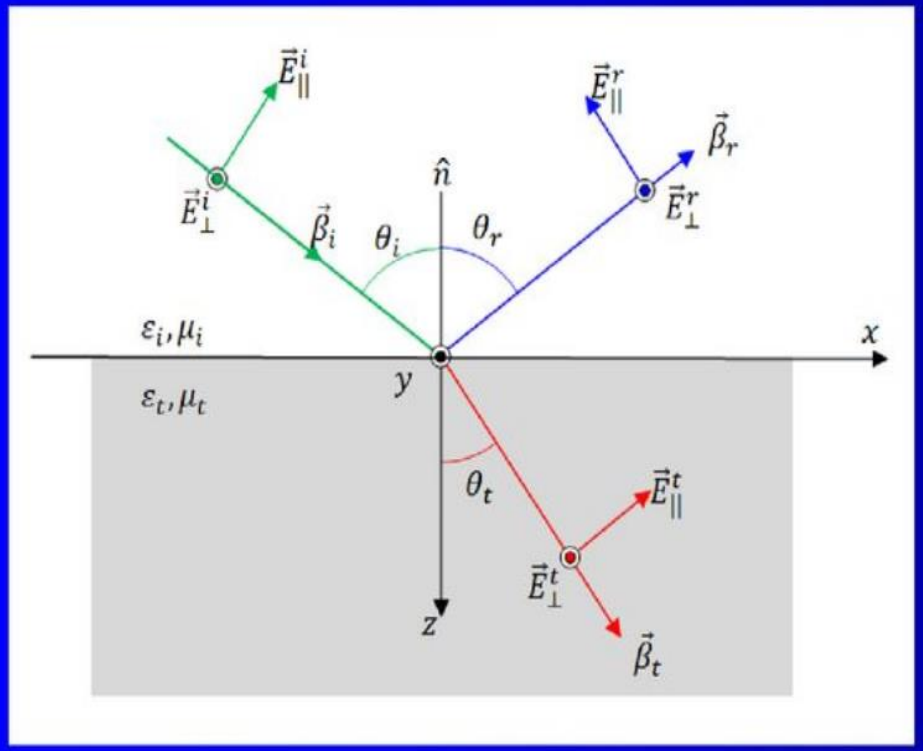
$$0° \leq \theta_i, \theta_r, \theta_t \leq 90°$$

$$\vec{E}_{\perp}^{t} = \vec{E}_{0\perp}^{t} e^{-j\vec{\beta}_t \cdot \vec{r}}$$

$$\vec{E}_{||}^{t} = \vec{E}_{0||}^{t} e^{-j\vec{\beta}_t \cdot \vec{r}}$$

Activa
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Reflection and Transmission Coefficients



$$\Gamma_{\perp} = \frac{E_{0\perp}^{r}}{E_{0\perp}^{i}} \qquad \Gamma_{\parallel} = -\frac{E_{0\parallel}^{r}}{E_{0\parallel}^{i}}$$

$$T_{\perp} = \frac{E_{0\perp}^{t}}{E_{0\perp}^{i}} \qquad T_{\parallel} = \frac{E_{0\parallel}^{t}}{E_{0\parallel}^{i}}$$

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Reflection and Transmission Coefficients

$$\Gamma_{\perp} = \frac{E_{0\perp}^{r}}{E_{0\perp}^{i}} = \frac{n_i \cos\theta_i - n_t \cos\theta_t}{n_i \cos\theta_i + n_t \cos\theta_t}$$

$$\Gamma_{||} = -\frac{E_{0||}^{r}}{E_{0||}^{i}} = -\frac{n_t \cos\theta_i - n_i \cos\theta_t}{n_t \cos\theta_i + n_i \cos\theta_t}$$

$$T_{\perp} = \frac{E_{0\perp}^{t}}{E_{0\perp}^{i}} = \frac{2n_i \cos\theta_i}{n_i \cos\theta_i + n_t \cos\theta_t}$$

$$T_{||} = \frac{E_{0||}^{t}}{E_{0||}^{i}} = \frac{2n_i \cos\theta_i}{n_t \cos\theta_i + n_i \cos\theta_t}$$

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Total (Internal) Reflection: Critical Angle

$$\Gamma_{\perp} = \frac{E_{0\perp}^{r}}{E_{0\perp}^{i}} = \frac{n_i \cos\theta_i - n_t \cos\theta_t}{n_i \cos\theta_i + n_t \cos\theta_t}$$

$$\Gamma_{||} = -\frac{E_{0||}^{r}}{E_{0||}^{i}} = -\frac{n_t \cos\theta_i - n_i \cos\theta_t}{n_t \cos\theta_i + n_i \cos\theta_t}$$

$\theta_i = \theta_c$ Critical angle

$\theta_t = 90°$ $\Gamma_{\perp} = 1$ $\Gamma_{||} = -1$

$$n_i \sin\theta_i = n_t \sin\theta_t$$

$$\sin\theta_c = n_t/n_i$$

$n_i > n_t$

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Total Transmission: Brewster Angle

$$\Gamma_{\perp} = \frac{E_{0\perp}^{r}}{E_{0\perp}^{i}} = \frac{n_i \cos\theta_i - n_t \cos\theta_t}{n_i \cos\theta_i + n_t \cos\theta_t}$$

$$\Gamma_{||} = -\frac{E_{0||}^{r}}{E_{0||}^{i}} = -\frac{n_t \cos\theta_i - n_i \cos\theta_t}{n_t \cos\theta_i + n_i \cos\theta_t}$$

$\theta_i = \theta_B$ Brewster angle (Polarization angle)

$\theta_B + \theta_t = 90°$ $\Gamma_{\perp} \neq 0$ $\Gamma_{||} = 0$

$n_i \sin\theta_i = n_t \sin\theta_t$

$n_i \sin\theta_B = n_t \cos\theta_B$

$\tan\theta_B = n_t/n_i$

Parallel component only

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